VOLUME 1

SOLUTION 0-2

Cicada 3301 - Liber Primus pages 0-2

27 x 27 rune matrix / three-rune node decryption

27 x 27 matrix / Euler totient / Mobius phase / Gematria Primus mod 29

SECTION 01

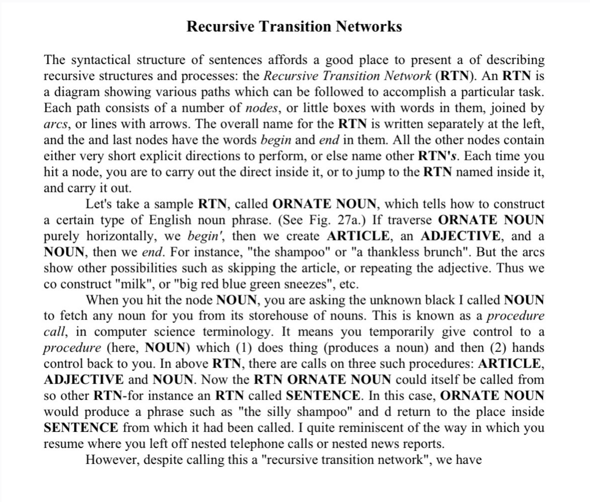
Opening clues before the matrix

Volume 1 begins with a small word construction before the 27 x 27 route.

```
ROAD = R(4) + O(3) + A(24) + D(23) = 54 -> 54 mod 29 = 25 -> AE
PATH = P(13) + A(24) + TH(2) = 39 -> 39 mod 29 = 10 -> I
WAY = W(7) + A(24) + Y(26) = 57 -> 57 mod 29 = 28 -> EA
```

The same key appears in the 3 x 3 matrix.

Pages 139-140 of GEB



Recursive Transition Networks

The syntactical structure of sentences affords a good place to present a of describing recursive structures and processes: the *Recursive Transition Network* (**RTN**). An **RTN** is a diagram showing various paths which can be followed to accomplish a particular task. Each path consists of a number of *nodes*, or little boxes with words in them, joined by *arcs*, or lines with arrows. The overall name for the **RTN** is written separately at the left, and the and last nodes have the words *begin* and *end* in them. All the other nodes contain either very short explicit directions to perform, or else name other **RTN'*s***. Each time you hit a node, you are to carry out the direct inside it, or to jump to the **RTN** named inside it, and carry it out.

Let's take a sample **RTN**, called **ORNATE NOUN**, which tells how to construct a certain type of English noun phrase. (See Fig. 27a.) If traverse **ORNATE NOUN** purely horizontally, we *begin'*, then we create **ARTICLE**, an **ADJECTIVE**, and a **NOUN**, then we *end*. For instance, "the shampoo" or "a thankless brunch". But the arcs show other possibilities such as skipping the article, or repeating the adjective. Thus we co construct "milk", or "big red blue green sneezes", etc.

When you hit the node **NOUN**, you are asking the unknown black I called **NOUN** to fetch any noun for you from its storehouse of nouns. This is known as a *procedure call*, in computer science terminology. It means you temporarily give control to a *procedure* (here, **NOUN**) which (1) does thing (produces a noun) and then (2) hands control back to you. In above **RTN**, there are calls on three such procedures: **ARTICLE**, **ADJECTIVE** and **NOUN**. Now the **RTN ORNATE NOUN** could itself be called from so other **RTN**-for instance an **RTN** called **SENTENCE**. In this case, **ORNATE NOUN** would produce a phrase such as "the silly shampoo" and d return to the place inside **SENTENCE** from which it had been called. I quite reminiscent of the way in which you resume where you left off nested telephone calls or nested news reports.

However, despite calling this a "recursive transition network", we have

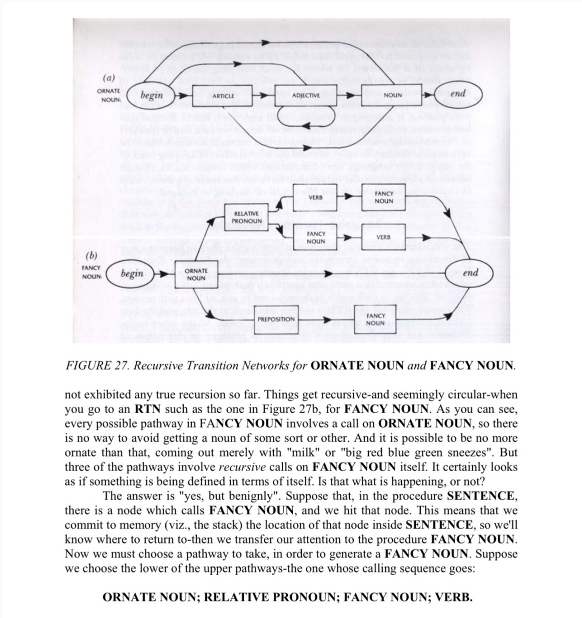


FIGURE 27. Recursive Transition Networks for **ORNATE NOUN** *and* **FANCY NOUN**.

not exhibited any true recursion so far. Things get recursive-and seemingly circular-when you go to an **RTN** such as the one in Figure 27b, for **FANCY NOUN**. As you can see, every possible pathway in FANCY NOUN involves a call on **ORNATE NOUN**, so there is no way to avoid getting a noun of some sort or other. And it is possible to be no more ornate than that, coming out merely with "milk" or "big red blue green sneezes". But three of the pathways involve *recursive* calls on **FANCY NOUN** itself. It certainly looks as if something is being defined in terms of itself. Is that what is happening, or not?

The answer is "yes, but benignly". Suppose that, in the procedure **SENTENCE**, there is a node which calls **FANCY NOUN**, and we hit that node. This means that we commit to memory (viz., the stack) the location of that node inside **SENTENCE**, so we'll know where to return to-then we transfer our attention to the procedure **FANCY NOUN**. Now we must choose a pathway to take, in order to generate a **FANCY NOUN**. Suppose we choose the lower of the upper pathways-the one whose calling sequence goes:

ORNATE NOUN; RELATIVE PRONOUN; FANCY NOUN; VERB.

Original Recursive Transition Networks reference preserved from Volume 1.

I’ve already used that part. Now add this to it:

> “believe nothing from this book”
> “except what you know to be true”
> “test the knowledge”

SECTION 02

What if the structure is the labyrinth?

I think the separators between the words may themselves be part of the clue. Another possibility is that the already decrypted message is meant to be hashed or checked through some underlying numerical pattern. In other words, I think the divisions may be an illusion, and that they are more likely serving as a kind of hash or checksum confirming the recovered plaintext - similar to the 233 relationship I found.

This could explain the strange word lengths that do not resemble natural English text. Linear reading is a false path. You cannot computationally crack something whose apparent structure is deliberately false from the start. That is why we need to think outside the box.

What if this entire thing is a huge map that deliberately disguises itself so that there is no single fixed movement formula?

Maybe there is only a core framework - phases, keys, and totients - and beyond that, you are expected to travel through the structure yourself, finding instructions and relationships as you go.

Right now, this interpretation is starting to make much more sense to me.

Maybe readers need to change their perspective entirely: what if there is no magical master key?

Even the wording in LP1 seems to point toward something like this:

> **LP1**
>
> EXPERIENCE YOUR DEATH
> TO THE GREAT JOURNEY
> IT IS NOT AN EASY TRIP
> ALONG THE WAY YOU WILL FIND AN END TO ALL STRUGGLE AND SUFFERING, YOUR INNOCENCE, YOUR ILLUSIONS, YOUR CERTAINTY

Do you see that final word - CERTAINTY?

SECTION 03

The working hypothesis

I am almost convinced that they constructed this manually so that the rules could change from one step to the next while the underlying structural core remained the same, forcing the solver to move deeper into the system while decrypting the message.

What do you think of that idea?

You naturally expect there to be some simple key that can be fed into a program and solved in five seconds. But what if they specifically did not want their work to be solved in five seconds? Or five minutes. Or five hours. Or five days. Or five months.

It could genuinely be a labyrinth - something that someone spent years constructing precisely because a structure like this would be extremely difficult to design in a short period of time.

And maybe they recruited people simply to help with the puzzle or with certain technical stages of it. But the core of the system was clearly designed by someone who must have spent at least a year working on it.

Would that person really want it to be solved in five seconds by some computer? I seriously doubt it.

And the text from LP1 seems to point in that direction as well - the journey, the loss of innocence, and the loss of certainty in anything.

PERSPECTIVE

So this kind of hand-built cipher may not have the kind of solution you are looking for at all. Maybe there is no single fixed formula that reproduces everything automatically. So I think we need to try changing our perspective.

Okay, let’s start from the beginning.

SECTION 04

Building the 27 x 27 map

We know for sure that 0-2 contains exactly 729 runes. That clearly forms a 27 x 27 matrix.

The simplest approach is to write the matrix from left to right, then move to the next row and continue reading from left to right.

This matrix stands out visually because you can see symmetry and reflection around the central axis. The drawings on 0-2 behave the same way, symmetrically, so this matrix is conceptually connected to 0-2.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	S/Z	H	E	O	G	M	IA/IO	F	S/Z	Y	E	NG	C/K	TH	J	G	AE	F	J	OE	O	N	TH	N	E	W	B
2	A	S/Z	OE	EO	I	NG	U/V	N	AE	G	I	TH	H	B	N	IA/IO	P	B	N	EO	J	TH	A	NG	Y	X	B
3	D	P	E	IA/IO	F	X	G	F	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	Y	H	B	U/V	G	O	P	NG
4	AE	TH	I	AE	H	X	TH	P	G	Y	F	NG	C/K	J	TH	A	M	I	NG	I	IA/IO	Y	H	EO	T	H	AE
5	IA/IO	X	Y	E	U/V	NG	C/K	F	N	EO	TH	J	I	TH	TH	P	Y	NG	EA	Y	N	E	Y	D	X	NG	W
6	B	C/K	X	D	B	F	M	T	N	E	EA	J	N	L	G	B	X	G	TH	Y	I	D	A	NG	G	M	J
7	R	O	L	EO	Y	T	P	I	TH	D	J	OE	H	L	EA	F	X	AE	P	G	X	F	L	W	EO	T	AE
8	F	G	N	M	EO	L	N	NG	M	F	R	IA/IO	A	S/Z	EA	M	NG	X	X	EO	D	B	OE	D	M	O	EA
9	EO	H	D	NG	G	OE	NG	L	U/V	R	N	T	AE	S/Z	Y	U/V	H	R	T	H	C/K	S/Z	OE	T	OE	NG	U/V
10	R	S/Z	F	O	E	W	EA	OE	L	F	AE	M	R	NG	D	A	M	R	W	A	I	M	S/Z	N	S/Z	OE	R
11	U/V	W	S/Z	L	IA/IO	OE	A	AE	NG	S/Z	D	P	T	C/K	R	TH	IA/IO	R	B	O	D	TH	F	Y	X	I	T
12	OE	I	F	L	O	F	EA	R	O	OE	M	H	M	G	P	H	AE	H	N	O	H	Y	OE	S/Z	L	N	G
13	AE	Y	OE	C/K	B	L	J	NG	L	OE	AE	J	EA	W	EA	TH	O	B	OE	Y	C/K	F	C/K	J	L	AE	N
14	TH	P	U/V	X	OE	X	G	P	F	S/Z	EO	AE	OE	NG	P	EO	O	E	A	G	AE	IA/IO	NG	TH	O	P	A
15	OE	TH	R	NG	AE	C/K	S/Z	B	EO	Y	H	T	X	E	L	R	G	W	Y	J	F	I	N	IA/IO	C/K	Y	EA
16	I	IA/IO	O	TH	E	C/K	AE	C/K	X	IA/IO	EA	O	L	C/K	F	R	D	NG	E	U/V	D	C/K	L	EA	S/Z	X	C/K
17	G	IA/IO	W	S/Z	TH	F	D	NG	I	IA/IO	M	A	AE	G	W	S/Z	N	D	C/K	P	TH	X	P	EA	EA	W	U/V
18	EA	W	F	W	J	Y	X	D	W	C/K	G	C/K	OE	D	X	OE	R	IA/IO	G	N	P	A	Y	P	C/K	Y	H
19	F	E	J	EA	N	OE	AE	U/V	A	H	R	E	EA	TH	F	J	A	L	X	S/Z	T	M	F	L	F	T	G
20	I	TH	U/V	L	G	X	EA	T	S/Z	L	J	P	L	X	I	OE	M	U/V	H	T	B	EO	L	D	H	B	M
21	EA	R	B	N	H	B	E	G	EO	D	L	W	EO	IA/IO	P	O	H	E	EA	W	I	R	I	H	U/V	TH	H
22	Y	N	X	B	G	J	P	U/V	NG	EA	F	N	I	E	D	IA/IO	NG	R	D	B	J	IA/IO	OE	M	I	F	T
23	J	D	I	TH	R	L	S/Z	E	EO	O	G	B	T	D	TH	A	N	C/K	Y	IA/IO	S/Z	TH	D	NG	F	L	E
24	G	H	C/K	E	O	I	T	N	X	P	T	F	H	R	D	E	F	T	AE	W	H	B	C/K	F	P	A	L
25	U/V	EA	N	L	J	J	C/K	L	W	EA	D	U/V	D	EO	EA	X	D	W	H	EA	TH	IA/IO	AE	C/K	L	T	W
26	E	I	C/K	P	OE	D	C/K	N	H	A	R	I	G	TH	EA	E	T	G	TH	H	O	IA/IO	W	AE	J	E	NG
27	EA	D	O	AE	A	L	B	J	C/K	U/V	X	T	A	B	J	P	F	EA	H	D	N	IA/IO	U/V	P	S/Z	U/V	W

The 729-rune 27 x 27 matrix in its initial form.

RANDOM-SHUFFLE TEST

I ran a test, and with random shuffling, the chance of getting exactly this kind of window next to the center NG of the 27 x 27 matrix is 0.0041%, or about 1 in 24,000.

SECTION 05

First node: AS I GO THE

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	S/Z	H	E	O	G	M	IA/IO	F	S/Z	Y	E	NG	C/K	TH	J	G	AE	F	J	OE	O	N	TH	N	E	W	B
2	A	S/Z	OE	EO	I	NG	U/V	N	AE	G	I	TH	H	B	N	IA/IO	P	B	N	EO	J	TH	A	NG	Y	X	B
3	D	P	E	IA/IO	F	X	G	F	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	Y	H	B	U/V	G	O	P	NG
4	AE	TH	I	AE	H	X	TH	P	G	Y	F	NG	C/K	J	TH	A	M	I	NG	I	IA/IO	Y	H	EO	T	H	AE
5	IA/IO	X	Y	E	U/V	NG	C/K	F	N	EO	TH	J	I	TH	TH	P	Y	NG	EA	Y	N	E	Y	D	X	NG	W
6	B	C/K	X	D	B	F	M	T	N	E	EA	J	N	L	G	B	X	G	TH	Y	I	D	A	NG	G	M	J
7	R	O	L	EO	Y	T	P	I	TH	D	J	OE	H	L	EA	F	X	AE	P	G	X	F	L	W	EO	T	AE
8	F	G	N	M	EO	L	N	NG	M	F	R	IA/IO	A	S/Z	EA	M	NG	X	X	EO	D	B	OE	D	M	O	EA
9	EO	H	D	NG	G	OE	NG	L	U/V	R	N	T	AE	S/Z	Y	U/V	H	R	T	H	C/K	S/Z	OE	T	OE	NG	U/V
10	R	S/Z	F	O	E	W	EA	OE	L	F	AE	M	R	NG	D	A	M	R	W	A	I	M	S/Z	N	S/Z	OE	R
11	U/V	W	S/Z	L	IA/IO	OE	A	AE	NG	S/Z	D	P	T	C/K	R	TH	IA/IO	R	B	O	D	TH	F	Y	X	I	T
12	OE	I	F	L	O	F	EA	R	O	OE	M	H	M	G	P	H	AE	H	N	O	H	Y	OE	S/Z	L	N	G
13	AE	Y	OE	C/K	B	L	J	NG	L	OE	AE	J	EA	W	EA	TH	O	B	OE	Y	C/K	F	C/K	J	L	AE	N
14	TH	P	U/V	X	OE	X	G	P	F	S/Z	EO	AE	OE	NG	P	EO	O	E	A	G	AE	IA/IO	NG	TH	O	P	A
15	OE	TH	R	NG	AE	C/K	S/Z	B	EO	Y	H	T	X	E	L	R	G	W	Y	J	F	I	N	IA/IO	C/K	Y	EA
16	I	IA/IO	O	TH	E	C/K	AE	C/K	X	IA/IO	EA	O	L	C/K	F	R	D	NG	E	U/V	D	C/K	L	EA	S/Z	X	C/K
17	G	IA/IO	W	S/Z	TH	F	D	NG	I	IA/IO	M	A	AE	G	W	S/Z	N	D	C/K	P	TH	X	P	EA	EA	W	U/V
18	EA	W	F	W	J	Y	X	D	W	C/K	G	C/K	OE	D	X	OE	R	IA/IO	G	N	P	A	Y	P	C/K	Y	H
19	F	E	J	EA	N	OE	AE	U/V	A	H	R	E	EA	TH	F	J	A	L	X	S/Z	T	M	F	L	F	T	G
20	I	TH	U/V	L	G	X	EA	T	S/Z	L	J	P	L	X	I	OE	M	U/V	H	T	B	EO	L	D	H	B	M
21	EA	R	B	N	H	B	E	G	EO	D	L	W	EO	IA/IO	P	O	H	E	EA	W	I	R	I	H	U/V	TH	H
22	Y	N	X	B	G	J	P	U/V	NG	EA	F	N	I	E	D	IA/IO	NG	R	D	B	J	IA/IO	OE	M	I	F	T
23	J	D	I	TH	R	L	S/Z	E	EO	O	G	B	T	D	TH	A	N	C/K	Y	IA/IO	S/Z	TH	D	NG	F	L	E
24	G	H	C/K	E	O	I	T	N	X	P	T	F	H	R	D	E	F	T	AE	W	H	B	C/K	F	P	A	L
25	U/V	EA	N	L	J	J	C/K	L	W	EA	D	U/V	D	EO	EA	X	D	W	H	EA	TH	IA/IO	AE	C/K	L	T	W
26	E	I	C/K	P	OE	D	C/K	N	H	A	R	I	G	TH	EA	E	T	G	TH	H	O	IA/IO	W	AE	J	E	NG
27	EA	D	O	AE	A	L	B	J	C/K	U/V	X	T	A	B	J	P	F	EA	H	D	N	IA/IO	U/V	P	S/Z	U/V	W

The first highlighted window and route near the matrix center.

```
Node: AE-J-EA
J=11 -> φ(11)=10=I
Key: AE-I-EA
Route: 10 + φ(10)=14 -> RIGHT 14
Encrypted runes: L-AE-N-TH-P-U-X
Key repeated: AE-I-EA-AE-I-EA-AE
Decrypt mod 29: L-AE-N-TH-P-U-X -> AS I GO THE
```

AS I GO THE

SECTION 06

One structure generates the next

This is the strongest part of the finding, because the key is simple, it matches exactly 7 runes, and it is directly linked to the totient of J.

```
AE-J-EA
-> J -> φ(J)=10=I
-> AE-I-EA
-> R14
-> 7 runes
-> AS I GO THE
```

```
final X
-> unique adjacent mirror X-OE-X
-> OE -> φ(OE)=10=I
-> X-I-X
-> RIGHT 10
-> exact center NG
-> 5 runes
-> X-X-I-X-X
-> WEATHER
```

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	S/Z	H	E	O	G	M	IA/IO	F	S/Z	Y	E	NG	C/K	TH	J	G	AE	F	J	OE	O	N	TH	N	E	W	B
2	A	S/Z	OE	EO	I	NG	U/V	N	AE	G	I	TH	H	B	N	IA/IO	P	B	N	EO	J	TH	A	NG	Y	X	B
3	D	P	E	IA/IO	F	X	G	F	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	Y	H	B	U/V	G	O	P	NG
4	AE	TH	I	AE	H	X	TH	P	G	Y	F	NG	C/K	J	TH	A	M	I	NG	I	IA/IO	Y	H	EO	T	H	AE
5	IA/IO	X	Y	E	U/V	NG	C/K	F	N	EO	TH	J	I	TH	TH	P	Y	NG	EA	Y	N	E	Y	D	X	NG	W
6	B	C/K	X	D	B	F	M	T	N	E	EA	J	N	L	G	B	X	G	TH	Y	I	D	A	NG	G	M	J
7	R	O	L	EO	Y	T	P	I	TH	D	J	OE	H	L	EA	F	X	AE	P	G	X	F	L	W	EO	T	AE
8	F	G	N	M	EO	L	N	NG	M	F	R	IA/IO	A	S/Z	EA	M	NG	X	X	EO	D	B	OE	D	M	O	EA
9	EO	H	D	NG	G	OE	NG	L	U/V	R	N	T	AE	S/Z	Y	U/V	H	R	T	H	C/K	S/Z	OE	T	OE	NG	U/V
10	R	S/Z	F	O	E	W	EA	OE	L	F	AE	M	R	NG	D	A	M	R	W	A	I	M	S/Z	N	S/Z	OE	R
11	U/V	W	S/Z	L	IA/IO	OE	A	AE	NG	S/Z	D	P	T	C/K	R	TH	IA/IO	R	B	O	D	TH	F	Y	X	I	T
12	OE	I	F	L	O	F	EA	R	O	OE	M	H	M	G	P	H	AE	H	N	O	H	Y	OE	S/Z	L	N	G
13	AE	Y	OE	C/K	B	L	J	NG	L	OE	AE	J	EA	W	EA	TH	O	B	OE	Y	C/K	F	C/K	J	L	AE	N
14	TH	P	U/V	X	OE	X	G	P	F	S/Z	EO	AE	OE	NG	P	EO	O	E	A	G	AE	IA/IO	NG	TH	O	P	A
15	OE	TH	R	NG	AE	C/K	S/Z	B	EO	Y	H	T	X	E	L	R	G	W	Y	J	F	I	N	IA/IO	C/K	Y	EA
16	I	IA/IO	O	TH	E	C/K	AE	C/K	X	IA/IO	EA	O	L	C/K	F	R	D	NG	E	U/V	D	C/K	L	EA	S/Z	X	C/K
17	G	IA/IO	W	S/Z	TH	F	D	NG	I	IA/IO	M	A	AE	G	W	S/Z	N	D	C/K	P	TH	X	P	EA	EA	W	U/V
18	EA	W	F	W	J	Y	X	D	W	C/K	G	C/K	OE	D	X	OE	R	IA/IO	G	N	P	A	Y	P	C/K	Y	H
19	F	E	J	EA	N	OE	AE	U/V	A	H	R	E	EA	TH	F	J	A	L	X	S/Z	T	M	F	L	F	T	G
20	I	TH	U/V	L	G	X	EA	T	S/Z	L	J	P	L	X	I	OE	M	U/V	H	T	B	EO	L	D	H	B	M
21	EA	R	B	N	H	B	E	G	EO	D	L	W	EO	IA/IO	P	O	H	E	EA	W	I	R	I	H	U/V	TH	H
22	Y	N	X	B	G	J	P	U/V	NG	EA	F	N	I	E	D	IA/IO	NG	R	D	B	J	IA/IO	OE	M	I	F	T
23	J	D	I	TH	R	L	S/Z	E	EO	O	G	B	T	D	TH	A	N	C/K	Y	IA/IO	S/Z	TH	D	NG	F	L	E
24	G	H	C/K	E	O	I	T	N	X	P	T	F	H	R	D	E	F	T	AE	W	H	B	C/K	F	P	A	L
25	U/V	EA	N	L	J	J	C/K	L	W	EA	D	U/V	D	EO	EA	X	D	W	H	EA	TH	IA/IO	AE	C/K	L	T	W
26	E	I	C/K	P	OE	D	C/K	N	H	A	R	I	G	TH	EA	E	T	G	TH	H	O	IA/IO	W	AE	J	E	NG
27	EA	D	O	AE	A	L	B	J	C/K	U/V	X	T	A	B	J	P	F	EA	H	D	N	IA/IO	U/V	P	S/Z	U/V	W

The AS I GO THE route feeding directly into the X-OE-X structure.

SECTION 07

Why WEATHER is structurally strong

Why I consider this strong:

1. The application of the AE-I-EA key ends exactly at the beginning of the mirrored structure X-OE-X.
2. The required shift is made from that exact starting point using φ(OE), where OE is the center of X-OE-X.
3. In the Gematria Primus alphabet used in Liber Primus, there are only two values whose totient equals 10: J and OE. And these are exactly the two central runes of the structures we encounter consecutively. That is already an extremely unlikely coincidence.
4. The next key begins exactly at NG, the center of the entire matrix, which is geometrically very elegant.
5. Directly above these ciphertext runes are exactly three runes: W-EA-TH, almost as if the authors were indicating how the phase should be chosen. And exactly three runes are sufficient to determine the phase unambiguously.

I found a direct rule for rotating the 3-rune key:

```
Phase = Σ μ(φ(Kᵢ)) mod 3
```

The result gives the starting position of the key:

```
0 = no rotation, 1 = start from rune 2, 2 = start from rune 3
Phase 0 -> start from rune 1
Phase 1 -> start from rune 2
Phase 2 -> start from rune 3
```

SECTION 08

Möbius phase: two consecutive keys

AE-I-EA:

```
φ(K) = 20, 4, 12
μ = 0, 0, 0
Σμ = 0
Phase = 0
-> AE-I-EA
```

X-I-X:

```
φ(K) = 6, 4, 6
μ = +1, 0, +1
Σμ = 2
Phase = 2
-> start from the 3rd rune
-> X-X-I
```

VERY STRONG CONNECTION

The probability of getting J as the center of one structure is 1/29, and the probability that the next structure has OE as its center is also 1/29. Therefore, the probability of the specific consecutive sequence J -> OE is $1/29^2 = 1/841 \approx 0.119\%$, or about 1 in 841. This is only the simplest estimate, before taking the other observed connections into account.

J and OE are the only values whose totient - the “loss of divinity” - equals 10, which is I.

And what does Liber Primus begin by discussing? Exactly: the nature of I.

That makes this feel very Cicada-like - hiding a thematic reference inside the mathematics itself.

SECTION 09

From WEATHER to the A crossroads

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	S/Z	H	E	O	G	M	IA/IO	F	S/Z	Y	E	NG	C/K	TH	J	G	AE	F	J	OE	O	N	TH	N	E	W	B
2	A	S/Z	OE	EO	I	NG	U/V	N	AE	G	I	TH	H	B	N	IA/IO	P	B	N	EO	J	TH	A	NG	Y	X	B
3	D	P	E	IA/IO	F	X	G	F	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	Y	H	B	U/V	G	O	P	NG
4	AE	TH	I	AE	H	X	TH	P	G	Y	F	NG	C/K	J	TH	A	M	I	NG	I	IA/IO	Y	H	EO	T	H	AE
5	IA/IO	X	Y	E	U/V	NG	C/K	F	N	EO	TH	J	I	TH	TH	P	Y	NG	EA	Y	N	E	Y	D	X	NG	W
6	B	C/K	X	D	B	F	M	T	N	E	EA	J	N	L	G	B	X	G	TH	Y	I	D	A	NG	G	M	J
7	R	O	L	EO	Y	T	P	I	TH	D	J	OE	H	L	EA	F	X	AE	P	G	X	F	L	W	EO	T	AE
8	F	G	N	M	EO	L	N	NG	M	F	R	IA/IO	A	S/Z	EA	M	NG	X	X	EO	D	B	OE	D	M	O	EA
9	EO	H	D	NG	G	OE	NG	L	U/V	R	N	T	AE	S/Z	Y	U/V	H	R	T	H	C/K	S/Z	OE	T	OE	NG	U/V
10	R	S/Z	F	O	E	W	EA	OE	L	F	AE	M	R	NG	D	A	M	R	W	A	I	M	S/Z	N	S/Z	OE	R
11	U/V	W	S/Z	L	IA/IO	OE	A	AE	NG	S/Z	D	P	T	C/K	R	TH	IA/IO	R	B	O	D	TH	F	Y	X	I	T
12	OE	I	F	L	O	F	EA	R	O	OE	M	H	M	G	P	H	AE	H	N	O	H	Y	OE	S/Z	L	N	G
13	AE	Y	OE	C/K	B	L	J	NG	L	OE	AE	J	EA	W	EA	TH	O	B	OE	Y	C/K	F	C/K	J	L	AE	N
14	TH	P	U/V	X	OE	X	G	P	F	S/Z	EO	AE	OE	NG	P	EO	O	E	A	G	AE	IA/IO	NG	TH	O	P	A
15	OE	TH	R	NG	AE	C/K	S/Z	B	EO	Y	H	T	X	E	L	R	G	W	Y	J	F	I	N	IA/IO	C/K	Y	EA
16	I	IA/IO	O	TH	E	C/K	AE	C/K	X	IA/IO	EA	O	L	C/K	F	R	D	NG	E	U/V	D	C/K	L	EA	S/Z	X	C/K
17	G	IA/IO	W	S/Z	TH	F	D	NG	I	IA/IO	M	A	AE	G	W	S/Z	N	D	C/K	P	TH	X	P	EA	EA	W	U/V
18	EA	W	F	W	J	Y	X	D	W	C/K	G	C/K	OE	D	X	OE	R	IA/IO	G	N	P	A	Y	P	C/K	Y	H
19	F	E	J	EA	N	OE	AE	U/V	A	H	R	E	EA	TH	F	J	A	L	X	S/Z	T	M	F	L	F	T	G
20	I	TH	U/V	L	G	X	EA	T	S/Z	L	J	P	L	X	I	OE	M	U/V	H	T	B	EO	L	D	H	B	M
21	EA	R	B	N	H	B	E	G	EO	D	L	W	EO	IA/IO	P	O	H	E	EA	W	I	R	I	H	U/V	TH	H
22	Y	N	X	B	G	J	P	U/V	NG	EA	F	N	I	E	D	IA/IO	NG	R	D	B	J	IA/IO	OE	M	I	F	T
23	J	D	I	TH	R	L	S/Z	E	EO	O	G	B	T	D	TH	A	N	C/K	Y	IA/IO	S/Z	TH	D	NG	F	L	E
24	G	H	C/K	E	O	I	T	N	X	P	T	F	H	R	D	E	F	T	AE	W	H	B	C/K	F	P	A	L
25	U/V	EA	N	L	J	J	C/K	L	W	EA	D	U/V	D	EO	EA	X	D	W	H	EA	TH	IA/IO	AE	C/K	L	T	W
26	E	I	C/K	P	OE	D	C/K	N	H	A	R	I	G	TH	EA	E	T	G	TH	H	O	IA/IO	W	AE	J	E	NG
27	EA	D	O	AE	A	L	B	J	C/K	U/V	X	T	A	B	J	P	F	EA	H	D	N	IA/IO	U/V	P	S/Z	U/V	W

The route after WEATHER, with the next highlighted structures around the central window.

WEATHER ends on E, which immediately begins the next structure: “EA-G-AE.” (only visually)

The continuation starts from A, effectively the left edge of that structure. This matches the earlier node pattern: AE-J-EA -> X-OE-X -> EA-G-AE.

Previously, J and OE reduced to I=10, and then: φ(I)=4.

So, if we simplify it, two things matter: 1. I and 2. the totient function. Therefore, it is logical to assume that the pattern around “I” may continue, and we should take the totient of I to get 4.

So we have two control signals, neither of them arbitrary: 1. I = 10 and 2. φ(I) = 4.

The EA-G-AE connection is just an elegant conceptual hint toward the previous structures. But it is still the same E-A-G-AE pattern, so starting from G is a false path - I found no continuation there.

If we continue from A right after WEATHER, the path appears immediately. I think EA-G-AE is simply a human-readable clue pointing to the same pattern - something a computer could easily miss, but a person would notice.

SECTION 10

Separators, dots, and the next two moves

I spent several hours building and checking this table by hand. From now on, we'll use only this map, since the separators may actually serve as markers or clues for how to move through it.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	S/Z	H	E	O	G	M	IA/IO	F 1	S/Z	Y	E	NG	C/K 13	TH	J	G	AE 1	F	J	OE 1	O	N	TH 1	N	E	W	B
2	A	S/Z	OE	EO	I	NG	U/V 1	N	AE	G	I	TH 1	H	B	N	IA/IO 1	P	B	N 1	EO	J	TH 1	A	NG	Y	X 1	B
3	D	P 1	E	IA/IO	F	X	G	F 1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO 1	Y	H 1	B	U/V 1	G	O	P 1	NG
4	AE	TH 1	I	AE	H	X	TH	P	G 1	Y	F	NG	C/K	J 1	TH	A	M	I	NG	I	IA/IO	Y 1	H	EO 1	T	H	AE
5	IA/IO 1	X	Y 1	E	U/V	NG 1	C/K	F	N 1	EO	TH	J	I	TH 1	TH	P 1	Y	NG	EA 1	Y	N	E	Y 1	D	X	NG	W 1
6	B	C/K	X	D	B	F 1	M	T	N	E 1	EA	J	N	L	G	B 1	X	G	TH 4	Y	I	D	A 1	NG	G	M	J
7	R	O	L	EO 1	Y	T	P	I	TH	D	J 1	OE	H	L 1	EA 1	F	X	AE	P	G	X 1	F	L	W	EO	T	AE 1
8	F	G	N 1	M	EO	L	N 1	NG	M	F	R	IA/IO 1	A	S/Z 1	EA	M	NG	X	X	EO	D	B 1	OE	D	M	O 1	EA
9	EO	H 1	D	NG	G 1	OE	NG	L	U/V	R	N	T 1	AE	S/Z	Y	U/V	H	R	T 1	H	C/K 1	S/Z	OE	T	OE	NG	U/V
10	R 1	S/Z 1	F	O	E	W	EA	OE	L	F	AE 1	M	R	NG 1	D	A	M	R 1	W	▶A	I	M	S/Z	N 1	S/Z	OE	R
11	U/V	W	S/Z	L	IA/IO 4	OE	A	AE	NG	S/Z	D	P	T 1	C/K	R	TH	IA/IO 1	R	B	O	D	TH	F 1	Y	X	I	T 4
12	OE	I 1	F	L	O 1	F	EA 1	R	O	OE	M	H	M	G	P	H 1	AE	H	N	O	H	Y 1	OE	S/Z	L 1	N	G 1
13	AE	Y 1	OE	C/K	B	L	J 1	NG	L	OE 1	AE	J	EA	W 1	EA	TH	O 1	B	OE	Y 1	C/K	F	C/K	J 1	L	AE 4	N 1
14	TH	P 1	U/V	X 1	OE	X	G 1	P	F	S/Z	EO	AE	OE 1	NG	P	EO	O	E	A 1	G	AE	IA/IO	NG	TH	O 1	P	A
15	OE	TH	R	NG	AE 1	C/K	S/Z	B	EO	Y	H 1	T	X	E	L	R 4	G	W	Y 1	J	F	I	N	IA/IO	C/K	Y	EA
16	I	IA/IO 1	O	TH 1	E	C/K	AE	C/K	X	IA/IO	EA 1	O	L	C/K 1	F	R	D	NG	E	U/V 1	D	C/K	L	EA	S/Z	X	C/K
17	G	IA/IO 4	W	S/Z	TH 1	F	D	NG 1	I	IA/IO	M	A	AE	G 1	W	S/Z 1	N	D 1	C/K	P	TH	X	P	EA	EA 1	W	U/V 1
18	EA	W 1	F	W	J	Y 1	X	D	W	C/K	G	C/K	OE 1	D	X	OE 1	R	IA/IO	G 1	N	P	A	Y	P 1	C/K	Y	H 1
19	F	E	J	EA	N 1	OE	AE 1	U/V	A 1	H	R 1	E	EA	TH	F	J	A 1	L	X	S/Z	T 1	M	F	L	F	T 1	G
20	I	TH 1	U/V	L	G 1	X	EA	T	S/Z	L	J	P 1	L	X	I	▶OE	M 1	U/V 4	H	T 1	B	EO	L	D	H	B	M 1
21	EA	R	B 1	N	H	B	E	G	EO 1	D	L	W	EO	IA/IO	P	O 1	H	E	EA 1	W	I	R	I	H 1	U/V	TH	H
22	Y 1	N	X	B	G	J	P	U/V 1	NG	EA	F	N	I	E	D	IA/IO	NG	R 1	D	B	J	IA/IO	OE	M	I 1	F	T 1
23	J	D	I	TH	R	L	S/Z 1	E	EO	O	G	B	T	D 1	TH	A	N	C/K	Y 1	IA/IO	S/Z	TH	D 1	NG	F	L	E
24	G	H	C/K 1	E	O	I	T	N	X 1	P	T	F	H	R	D	E	F	T 1	AE	W	H 1	B	C/K 1	F 1	P	A	L
25	U/V	EA	N	L	J 1	J	C/K	L	W	EA	D	U/V	D	EO 1	EA	X	D	W	H	EA 1	TH	IA/IO	AE	C/K	L	T	W
26	E	I	C/K 1	P	OE	D	C/K 1	N	H	A 1	R	I	G	TH	EA	E	T	G 1	TH	H	O	IA/IO	W	AE	J	E 1	NG
27	EA	D 1	O	AE 1	A	L 1	B	J	C/K	U/V	X	T	A	B	J	P 1	F	EA 1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W 13

The hand-checked map with the route and separator pattern visible together.

Here are the strongest and most unusual things I've found:

1. AS I GO THE fits the separators perfectly, which may be a clue that the starting point is correct.
2. After WEATHER, the pattern seems to continue for one more cell before the separator appears, effectively pointing directly to A (14,19). This may be a clue that A is the intended next point.
3. Around I, there are 4 and 1 dots, and φ(I)=4, which is an interesting observation.

Also, look at the overall structure of the dots themselves. They form either very clear diagonals or distinct vertical lines, for example in column 2.

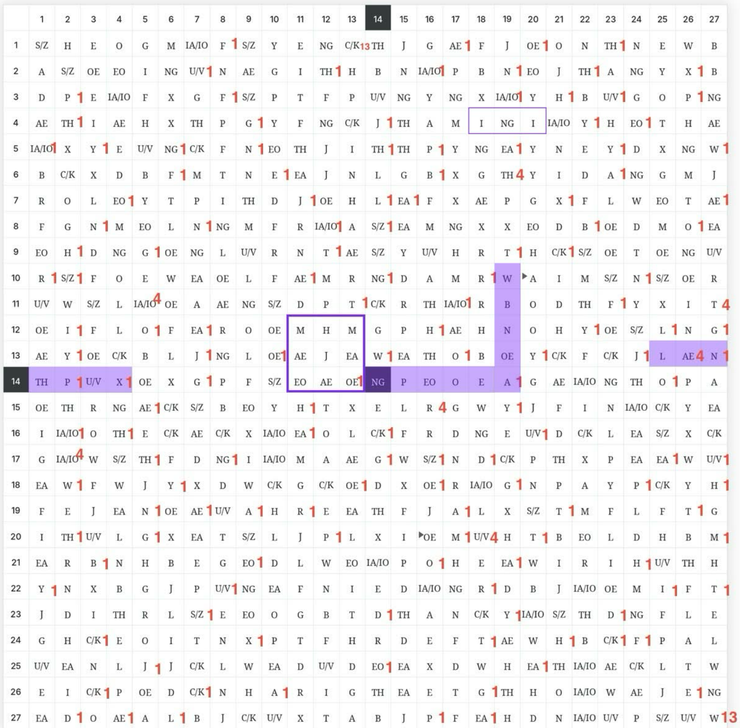
From AE-J-EA and X-OE-X, both J and OE reduce to I=10 through the totient function. Applying totient once more gives φ(I)=4, so the two movement values are 10 and 4.

A CROSSROADS

After WEATHER, A(14,19) acts as a crossroads: moving 10 up reaches NG, and moving 4 right reaches another NG. This defines the route UP -> RIGHT.

SECTION 11

The H-NG-C key and TURNS



The A crossroads and the vertical structure reached from the right-hand NG.

From the right NG, moving 10 up gives H and 10 down gives C, forming the key H-NG-C. Its totient pattern is 4-12-4, the same as I-NG-I.

Reading upward from A gives A-OE-N-B-W, and subtracting the repeating key H-NG-C-H-NG gives TURNS.

```
Ciphertext: A-OE-N-B-W
Key: H-NG-C-H-NG
Plaintext: T-U-R-N-S
```

TURNS

The key H-NG-C can be seen as the vertical axis of the same cross, rather than something chosen separately.

UNIQUENESS

What makes this especially interesting is that, in the entire 27 x 27 matrix, NG(14,23) is the only NG with A exactly 4 cells away on both the left and the right.

SECTION 12

What lies below the crossroads?

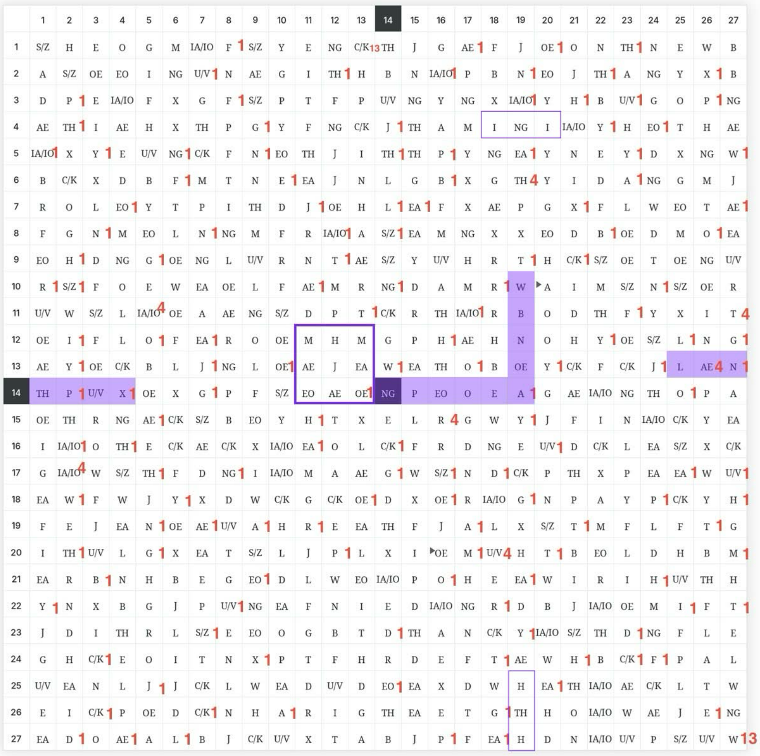
```
              H (4,23)
                 |
                10
                 |
A (14,19) --4-- NG (14,23) --4-- A (14,27)
                 |
                10
                 |
              C (24,23)
```

Cross geometry around A(14,19) and NG(14,23).

Since we are using the cleanest approach, the next step should be simple and logical.

Above A, we found I-NG-I, which deserves further study. But what lies below?

Below, we find H-TH-H - the only mirrored structure in this lower area, both vertically and horizontally. The vertical occurrence stands out much more clearly because it lies entirely within A's column. This makes H-TH-H the natural next key point.



The larger route with the lower H-TH-H structure incorporated.

I also noticed that WAY passes through A, our highly connected intersection, with A at its center. This may be a hint from the authors that we are on the right path.

SECTION 13

WEATHER: the compact case

I have strong arguments for WEATHER, which I already mentioned above.

1. The key-phase rule works perfectly with the X-OE-X structure.
2. AS I GO THE ends exactly on X-OE-X.
3. So one structure generates the next.
4. The shift by 10 is not arbitrary - it is a shift by I, because in Primus gematria I = 10.
5. Why I? Because AS I GO THE literally points to I. The totients of OE and J are the only ones that give I. So it makes sense that the result of observation of the AS I GO THE structure is then used as the shift to find the next one. And shifting by exactly 10 lands on NG, the center of the 27 x 27 matrix, where the word begins.
6. Directly beneath WEATHER are three runes that hint at the phase-selection rule: W EA TH.

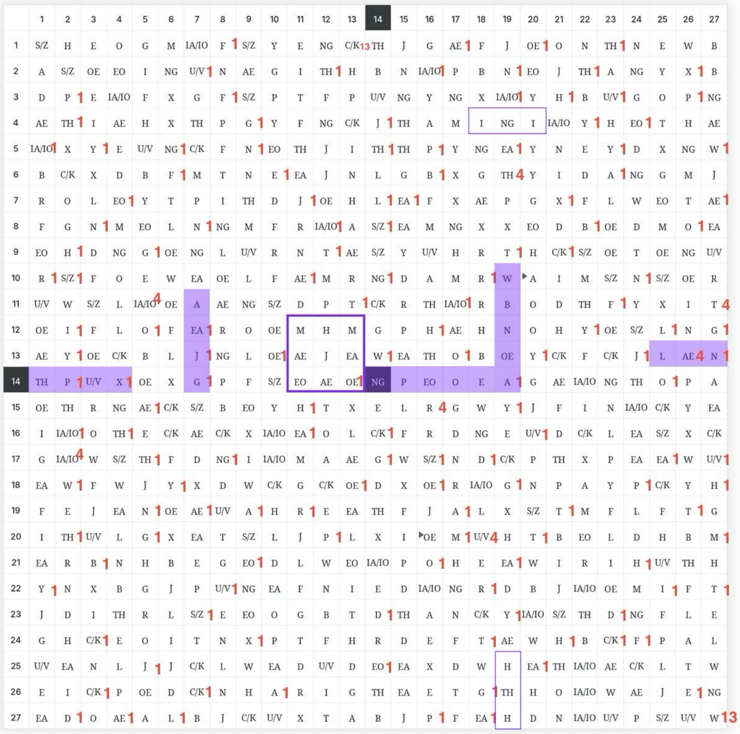
IN ONE LINE

Literally: shift by I, and you land in the heart of the solution. That feels very Cicada-like.

Put all of this together, and you get the next stage. Try thinking outside the box - you’ll notice plenty of references to “I” too.

SECTION 14

The NG value 12 links ciphertext and key



The extended route as the 12-step branch is introduced.

While examining the A crossroads, we found that two different directions from A lead to NG centers, and both resulting structures share the same totient signature.

Now comes the interesting part.

```
φ(NG) = φ(21) = 12
```

Exactly 12 cells down from A lands on the center of H-TH-H.

And exactly 12 cells left from A lands on G(14,7), which is the starting point of the ciphertext for the next word.

The key for that ciphertext is derived from the structure reached by the other 12-step branch: H-TH-H -> H-U-H.

ONE VALUE, TWO TARGETS

So the NG discovered at the previous stage does not merely end one step. Through its totient value 12, it simultaneously points to: 1. the next ciphertext start; 2. the structure that generates its key. One value derived from the previous NG connects the next plaintext and its key at the same time.

SECTION 15

Decrypting COLD

Node compilation:

```
K = (a, φ(b), c)
H-TH-H -> H-U-H
because φ(TH=2)=1=U
```

Phase:

```
p = [μ(φ(k1)) + μ(φ(k2)) + μ(φ(k3))] mod 3
For H-U-H:
p = [μ(4) + μ(1) + μ(4)] mod 3
= [0 + 1 + 0] mod 3
= 1
So the active key is: H-U-H -> U-H-H
```

Ciphertext:

```
G-J-EA-A
```

Decryption:

```
P = C - K mod 29
G-J-EA-A - U-H-H-U = C-O-L-D
```

COLD

The discovered rule for selecting the key phase worked perfectly.

> “AS I GO, THE WEATHER TURNS COLD”

SECTION 16

Numerical coherence around the phrase

Within the phrase “AS I GO, THE WEATHER TURNS COLD,” several numerical relationships line up in an unusually coherent way.

```
7 words
21 runes -> 21 = 3 x 7
sum of 0-based GP indices = 233
COLD index sum = 51
233 is the 51st prime number
NG value = 21 and NG is at the matrix center
F7 = 13 and F13 = 233
COLD sits exactly in column 7
```

So the numbers 7, 21, 51, and 233 connect the phrase through word count, rune count, Gematria Primus, the matrix center, prime numbers, and Fibonacci structure.

I also noticed that COLD sits exactly in column 7. It all ties together so elegantly.

Column 19 seems especially important because several key structures are anchored to it: the A crossroads, I-NG-I, and H-TH-H. This already suggests that 19 was chosen deliberately.

The numerical link makes that even more striking: W has a zero-based GP index of 7 and a prime value of 19, and W is the only rune for which the difference between these two values is exactly 12:

```
19 - 7 = 12 = φ(NG)
```

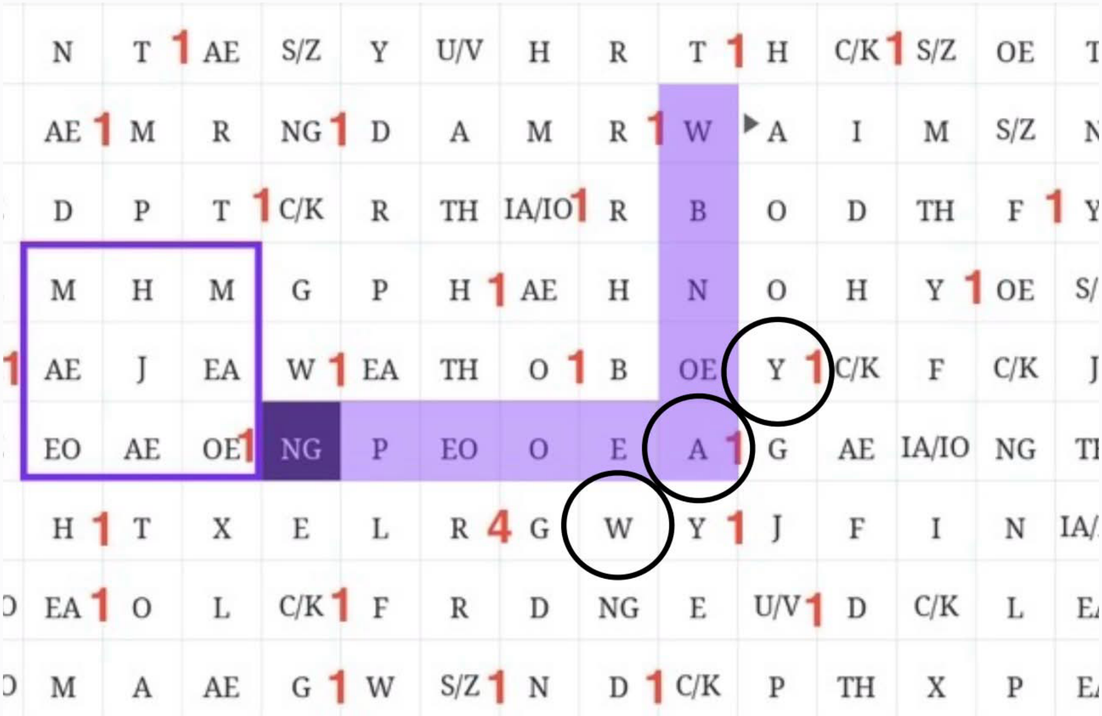
So the independently derived NG shift connects 19 directly to 7. The geometry follows the same pattern: TURNS ends on W in column 19, while COLD begins in column 7. This makes column 19 look like a deliberate transition point, with φ(NG)=12 providing the exact move from 19 to 7. Profoundly beautiful.

SECTION 17

The final cell keeps pointing forward

The final ciphertext cell of each stage seems to hint at the next step.

```
AS I GO THE ends on X -> opens X-OE-X
WEATHER ends at the E/A boundary -> leads into the A crossroads
TURNS ends on W in column 19 -> points to the 19 -> 7 transition via φ(NG)=12
COLD ends on A(11,7) -> φ(A)=8=H, reinforcing H-TH-H
```



A close view of the densely connected crossroads and surrounding structures.

God, Cicada were brilliant. I didn’t even notice this at first. A is literally a crossroads - a point where roads intersect or split.

And at the end of LP2, there is the final instruction:

LP2

FIND THE DIVINITY WITHIN AND EMERGE

```
DIVINITY WITHIN = 491
“A” CROSSROADS = 491
```

That is such a beautiful piece of wordplay. They placed it right in plain sight. Even the directional symbols drawn on pages 0-2 - which are not simply crosses, but markers indicating possible directions - fit the same idea conceptually: they resemble crossroads or branching paths.

SECTION 18

Prime and separator side-observations

Interestingly, message.txt.asc from 2014 contained a hidden sequence of spaces that encoded every prime number from 2 to 37 - all except the prime number 19.

And in my solution, column 19 plays a key role, as also does the 19 -> 7 transition. The choice of 37 is obviously a play on 3 and 7.

if you add up all the primes from “as i go the” sums up to 301. (like 3301)

Good find. I hadn’t noticed that. It may be intentional, since it resembles 3301 (301). I haven’t focused much on primes, though, since indices are easier for encoding clues.

Also, the separators across all 7 words sum to 14:

```
1+4+1+1+1 +1+1+1 +1+1+1 = 14
```

AS I GO, THE WEATHER TURNS COLD.